Kai S. Yun

Legal Name: SirkHoo Yun
Ph.D. Student, MIT | Laboratory for Information & Decision Systems (LIDS)
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RESEARCH INTEREST

Leveraging the intersection of **control theory**, **deep learning**, and **foundation models** to develop and deploy **safe, scalable controllers** on **physical robots**, such as humanoids, quadrupeds, and drones.

EDUCATION

Massachusetts Institute of Technology — Class of 2029 (Expected)
Ph.D. in Mechanical Engineering — Cambridge, MA

Carnegie Mellon University — Class of 2025
M.S. in Mechanical Engineering — Pittsburgh, PA

University of California, Berkeley — Class of 2023
B.S. in Mechanical Engineering, Minor in EECS — Berkeley, CA

HONORS & AWARDS

NeurIPS Spotlight Paper Selection (Top 1.3%) [P1], 2026
MathWorks Fellowship, 2026 — 2027, MIT School of Engineering
ATK-Nick G. Vlahakis Fellowship, 2024 — 2025, CMU College of Engineering
Best Master's Research Paper Commencement Award, 2025, CMU Department of Mechanical Engineering
Conference Grants: MIT OGE (2026), CMU OGPA (2024), CMU ME (2024)

PUBLICATIONS

P: preprints, C: conference proceedings, J: journal articles, *equal contribution

[P1] **Kai S. Yun**, Zeyang Li, Navid Azizan, "Provably Safe, Yet Scalable Reinforcement Learning." Accepted to *NeurIPS 2026 (**Spotlight**)*. Preprint at *arxiv.org:2606.14536*, 2026.

[C2] **Kai S. Yun**, Navid Azizan, "ATOM-CBF: Adaptive Safe Perception-Based Control under Out-of-Distribution Measurements." Published in *Learning for Dynamics and Control Conference (L4DC)*, 2026. Poster Presentation in Los Angeles, California, 2026. Full paper at *arxiv.org:2511.08741*.

[C3] Yifan Sun, Rui Chen, **Kai S. Yun**, Yikuan Fang, Sebin Jung, Feihan Li, Bowei Li, Weiye Zhao, Changliu Liu, "SPARK: Safe Protective and Assistive Robot Kit." Published in *IFAC Symposium on Robotics*, 2025. Full paper at *arxiv.org:2502.03132*, 2025, **Website**.

[C4] **Kai S. Yun**, Rui Chen, Chase Dunaway, John M. Dolan, Changliu Liu, "Safe Control of Quadruped in Varying Dynamics via Safety Index Adaptation." Published in *International Conference on Robotics and Automation (ICRA)*, 2025. Oral Presentation in Atlanta, Georgia, 2025, **Website**.

[C5] Simin Liu*, **Kai S. Yun***, John M. Dolan, Changliu Liu, "Synthesis and Verification of Robust-Adaptive Safe Controllers." Published in *European Control Conference (ECC)*, 2024. Oral Presentation in Stockholm, Sweden, 2024.

[C6] Tianhao Wei, Hanjiang Hu*, Luca Marzari*, **Kai S. Yun***, Peizhi Niu*, Xusheng Luo, Changliu Liu, "ModelVerification.jl: a Comprehensive Toolbox for Formally Verifying Deep Neural Networks." Published in *International Conference on Computer Aided Verification (CAV)*, 2025, **GitHub**.

[J7] Jack C. Harms, Ethan M. Grame, **SirkHoo Yun**, Bushra Ahmed, Leah C. O'Brien, James J. O'Brien, "Mass-independent Dunham Analysis of the $[7.7]Y^2\Sigma^+ - X^2\Pi_i$ and $[16.3]A^2\Sigma^- - X^2\Pi_i$ Transitions of Copper Monoxide, CuO." Published in *Journal of Molecular Spectroscopy*, 2019.

RESEARCH EXPERIENCE

Laboratory for Information & Decision Systems (MIT) Aug 2025 — Present
Advisor: Prof. Navid Azizan Cambridge, MA

- Vision-language-action (VLA) models for safe, scalable, and general-purpose robotics.
- Hard-constrained safe reinforcement learning for high-dimensional robots [P1].
- Perception-based safe control of robots for out-of-distribution measurements [C2].

Robotics Institute (CMU) Jun 2023 — Jul 2025
Advisors: Prof. Changliu Liu, Prof. John M. Dolan Pittsburgh, PA

- Safe humanoid autonomy and teleoperation toolbox (*SPARK*) (Unitree G1 platform) [C3].
- Adaptive safe controller for quadrupeds in varying dynamics (Unitree Go2 platform) [C4].
- Algorithm for synthesizing robust-adaptive safe controllers under uncertainty [C5].
- *ModelVerification.jl*, a Julia library containing state-of-the-art DNN verification algorithms [C6].

Berkeley AI Research Lab (UC Berkeley) Aug 2021 — Apr 2022
Advisor: Prof. Koushil Sreenath Berkeley, CA

- *PPO-Worst-Case*, a novel safe reinforcement learning algorithm for worst-case scenarios.
- Simulations for legged robots in safety-critical scenarios.

PROFESSIONAL EXPERIENCE

Tesla, Inc. May 2022 — Aug 2022
Vehicle Dynamics / Software Engineering Intern Palo Alto, CA

- Developed a framework for comparing subjective vehicle evaluation and objective test data.
- Tested and analyzed vehicle dynamics performance of Models S, 3, and Semitruck.
- Developed automated internal ticketing, reporting, and logging systems.

NeuroCore.ai Oct 2020 — Jul 2021
Reinforcement Learning Engineering Intern Seoul, South Korea

- Designed and developed RL training and deployment frameworks; 84% increase in training efficiency.
- Developed simulators for Supply Chain Management (SCM) tasks for semiconductor manufacturers.
- These are currently deployed in South Korean semiconductor manufacturers, such as SK Hynix.

SKILLS

Expertise: Humanoid, Quadruped, Drone, ML/RL, VLA, World Model, Safe Control, Optimization
Physical Robots/Machines: Unitree G1, Go2, Pixhawk Drones, Sawyer, F1Tenth, Tanks
Libraries: JAX, PyTorch, NumPy, MJX, Gazebo, Isaac, PyBullet, Mosek, CasADi, PX4
Programming/OS/Tools: Python, MATLAB, Julia, C++, ROS, ROS2, Linux, Git
Other skills: Jetson, OptiTrack/Vicon, Simulink, Scuba Diving
Languages: Fluent in English & Korean

TEACHING & SERVICE

Reviewer Service

Transactions on Machine Learning Research (TMLR)
IEEE Robotics and Automation Letters (RA-L)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Robotics Institute at CMU Pittsburgh, PA

Manager/Developer — RoboticsKnowledgebase (Funded by Google) Oct 2024 — Jul 2025
Teaching Assistant — F1Tenth Autonomous Racing (16.663) Jan 2025 — May 2025
Robot Manager, Lab Manager — Intelligent Control Lab Jan 2025 — May 2025
Lab Server Manager — DRIVE Lab Jun 2024 — May 2025
Research Mentor — Robotics Institute Summer Scholars (RISS) May 2024 — Aug 2024
Mentee: Chase Dunaway (Undergraduate at New Mexico Tech → Ph.D. at MIT)

Mechanical Engineering Department at CMU Pittsburgh, PA

Teaching Assistant — Dynamic Systems and Controls (24.352) Aug 2024 — Dec 2024

Mechanical Engineering Department at UC Berkeley Berkeley, CA

Teaching Assistant — Statistics and Data Science for Engineers (ENG 178) Jan 2023 — May 2023

Republic of Korea Army South Korea

Squad Leader, Sergeant, Tank Mechanic — Combat Service Support (CSS) Jan 2019 — Aug 2020

WORK AUTHORIZATION

US Permanent Resident (Green Card holder), South Korean Citizen (military service completed)